Comparative Analysis of LSTM-Attention and Temporal Fusion Transformer for Stock Forecasting

Haswika
Department of Electrical Engineering
Indian Institute of Technology Dharwad
Email: ee23bt020@iitdh.ac.in

***Abstract*—Forecasting stock prices is a challenging time-series prediction problem due to nonlinear dependencies, sudden fluctuations, and market volatility. While recurrent neural networks such as LSTMs have shown promising performance, transformer-based models have recently demonstrated superior forecasting ability through attention mechanisms and dynamic feature selection. In this work, we compare an LSTM-Attention model and a Temporal Fusion Transformer (TFT) for multi-step stock forecasting on two datasets with contrasting volatility characteristics: Tesla (TSLA) and the S&P 500 index. A hybrid ensemble combining both models is also evaluated. Experimental results show that TFT consistently outperforms LSTM-Attention, while the ensemble achieves the best overall RMSE on both datasets. Further analysis shows that volatility-sensitive features are more influential for highly volatile assets such as TSLA. These results demonstrate the effectiveness of transformer-based architectures for financial forecasting tasks.**

***Index Terms*—Stock forecasting, LSTM-Attention, Temporal Fusion Transformer, Ensemble learning, Time-series prediction**

I. Introduction

Stock price prediction has been an important research problem in financial analytics due to its practical relevance in investment decision-making and risk management. However, forecasting financial time series is difficult because stock prices exhibit strong nonlinearity, noise, and high sensitivity to market events.

Traditional statistical approaches such as ARIMA are effective for linear and stationary time-series data but are limited in their ability to capture complex temporal dependencies. Deep learning models such as Long Short-Term Memory (LSTM) networks improve performance by learning sequential relationships over time. More recently, transformer-based architectures have shown strong forecasting capability by incorporating attention mechanisms that capture long-range dependencies more effectively.

This work compares the forecasting performance of LSTM-Attention and Temporal Fusion Transformer (TFT) models on two datasets with distinct volatility profiles: Tesla (TSLA), representing a highly volatile stock, and the S&P 500 index, representing a relatively stable benchmark. A hybrid ensemble model is also proposed to combine the strengths of both architectures.

II. Related Work

LSTM-based models have been widely used for stock forecasting due to their ability to model sequential dependencies in financial data. Several studies have demonstrated that LSTMs outperform traditional machine learning methods in predicting short-term price movements. However, standard LSTMs often struggle with long-range dependencies and may not effectively identify the most relevant historical patterns.

Attention mechanisms improve recurrent models by assigning dynamic importance weights to past observations, enabling the model to focus on informative timesteps. Transformer-based models extend this idea through self-attention, allowing parallel modeling of long-range dependencies.

The Temporal Fusion Transformer (TFT) combines recurrent layers, variable selection networks, and interpretable attention mechanisms, making it highly suitable for multi-horizon forecasting tasks. In addition, explainability methods such as SHAP have been increasingly used in financial prediction models to identify the relative importance of input features.

III. Dataset and Feature Engineering

Historical stock data for TSLA and the S&P 500 index was collected from January 2019 to December 2024 using the `yfinance` library. The raw features included Open, High, Low, Close, and Volume values.

To improve predictive performance, multiple technical indicators were generated. Trend indicators included Simple Moving Average (SMA), Exponential Moving Average (EMA), and MACD. Momentum indicators included RSI, stochastic oscillator, and rate of change (ROC). Volatility indicators such as Bollinger Band Width and Average True Range (ATR) were also included, along with lag features based on historical closing prices.

A total of 23 engineered features were used for both datasets. To preserve temporal order and prevent data leakage, the datasets were divided using a time-based split of 70% training, 15% validation, and 15% testing.

IV. Methodology

A. LSTM-Attention Model

The LSTM-Attention model takes a 60-day sequence of engineered features as input and predicts the next five closing prices. A three-layer LSTM network with 256 hidden units was used to encode temporal information.

To improve interpretability and forecasting performance, an attention mechanism was applied over the hidden states.

Instead of relying solely on the final hidden state, the attention layer learns weights for all timesteps, allowing the model to focus on the most relevant portions of the input sequence. The weighted representation is then passed through a fully connected layer to generate the final multi-step prediction.

B. Temporal Fusion Transformer

The Temporal Fusion Transformer (TFT) was selected due to its effectiveness in multi-horizon forecasting. TFT combines recurrent processing with variable selection networks, gated residual connections, and multi-head self-attention.

The variable selection network identifies the most relevant features at each timestep, improving interpretability and reducing noise from less useful inputs. Multi-head self-attention captures long-range dependencies, allowing the model to identify patterns beyond the local context. TFT also predicts quantiles, enabling uncertainty estimation through prediction intervals.

C. Hybrid Ensemble

To combine the strengths of both models, a weighted ensemble was implemented as:

$$\hat{y}_{ens} = 0.3\hat{y}_{LSTM} + 0.7\hat{y}_{TFT} \tag{1}$$

where the TFT output receives a higher weight due to its stronger validation performance.

V. Results and Discussion

The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Directional Accuracy (DA).

TABLE I
PERFORMANCE COMPARISON ON TSLA

Model	MAE	RMSE	MAPE	DA
LSTM-Attention	14.19	21.36	5.15	53.88
TFT	13.69	19.76	5.57	52.33
Ensemble	**9.89**	**13.09**	**4.95**	50.65

TABLE II
PERFORMANCE COMPARISON ON S&P 500

Model	MAE	RMSE	MAPE	DA
LSTM-Attention	69.56	92.27	1.23	47.29
TFT	61.26	83.22	1.11	55.77
Ensemble	**62.59**	**82.54**	1.17	54.78

The TFT model outperformed the LSTM-Attention model on both datasets, particularly on the S&P 500 index. This improvement is attributed to TFT's ability to perform dynamic feature selection and model long-term dependencies more effectively.

The ensemble model achieved the lowest RMSE for both TSLA and S&P 500, demonstrating that combining LSTM-Attention and TFT predictions improves robustness. While

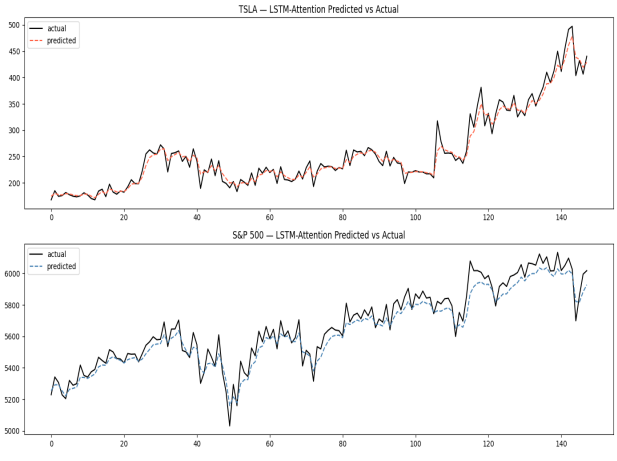


Fig. 1. Actual vs predicted prices using lstm.

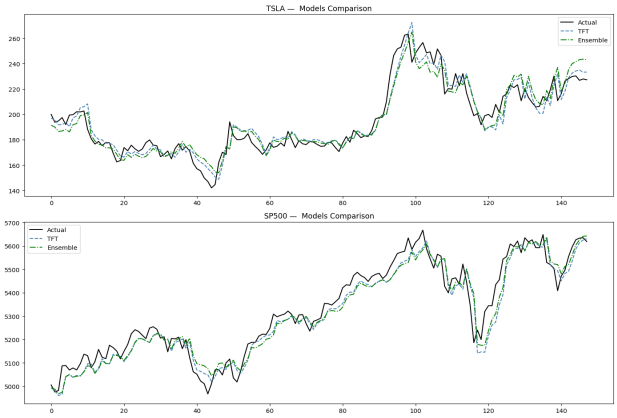


Fig. 2. Actual vs predicted prices using the hybrid ensemble model and tft.

LSTM captures short-term sequential trends effectively, TFT contributes stronger global forecasting capability.

The prediction intervals generated by TFT were wider for TSLA compared to S&P 500, indicating greater uncertainty due to higher market volatility.

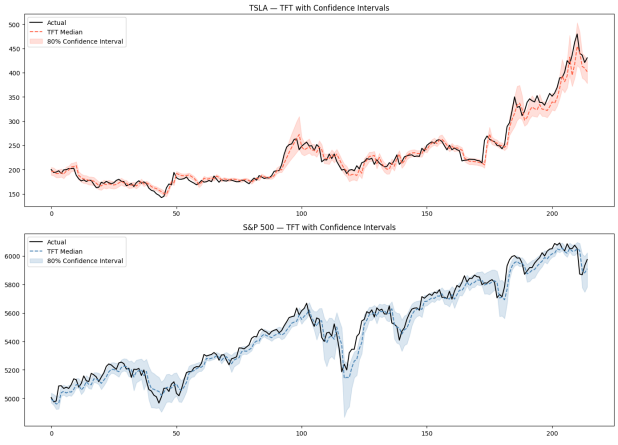


Fig. 3. TFT forecasts with confidence intervals for TSLA and S&P 500.

Feature importance analysis showed that momentum indicators such as MACD difference and stochastic oscillator were important for both datasets, while volatility-related features had greater importance for TSLA.

VI. Conclusion

This work compared LSTM-Attention and Temporal Fusion Transformer models for multi-step stock forecasting on volatile and stable market datasets. TFT achieved better standalone performance than LSTM-Attention, while the hybrid ensemble produced the best overall RMSE. The results also highlight that volatility-related indicators are more important for forecasting highly volatile stocks such as TSLA.

Future work may incorporate sentiment analysis from financial news and test the framework across a broader range of stocks and sectors.

References

[1] B. Lim et al., "Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting," *International Journal of Forecasting*, 2021.

[2] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, 1997.

[3] D. Bahdanau et al., "Neural Machine Translation by Jointly Learning to Align and Translate," *ICLR*, 2015.

[4] S. Lundberg and S. Lee, "A Unified Approach to Interpreting Model Predictions," *NeurIPS*, 2017.